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08/03/2026



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AUDIO DIGITAL PROCESSING & INTELLIGENCE GENRE DETECT SYSTEM

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Introduction

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Project Objectives

- Design a multi-band Digital Equalizer based on FFT.
- Implement real-time audio processing using Web Audio API.
- Extract features for music genre classification.
- Train and deploy a Random Forest classifier.
- Visualize waveform and frequency spectrum before/after processing.
- Build a complete Frontend–Backend web application.



Scope

- Web-based application (React + FastAPI).
- Offline FFT-based Equalizer (backend).
- Realtime IIR-based Equalizer (browser).
- Supervised music genre classification.
- Mono audio processing.

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Architecture

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Visualize

EQ (Offline Mode):

- Waveform and spectrum for both the original signal and the processed signal after EQ are computed in the backend.
- The results are returned via API and visualized on the frontend.

Realtime Mode:

- Audio is processed in continuous frames within the DSP pipeline.
- Waveform and spectrum are captured from frames before and after EQ processing, and updated continuously for realtime visualization.

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EQ File Processing

Uploaded / Recorded Audio File

→ Load & Convert to Mono (librosa)

→ Frame Processing

→ FFT

→ Apply Multi-Band Gain

→ IFFT

→ Overlap-Add

→ Normalize Signal

→ Waveform & Spectrum Visualization

→ Export Processed Audio

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EQ Realtime Processing

Band	Tần số	Vai trò
Bass	20–180 Hz	tiếng trống, bass
Low-mid	180–450 Hz	độ dày của nhạc
Mid	450–2200 Hz	giọng hát
High-mid	2200–5000 Hz	độ rõ
Treble	5000–20000 Hz	cymbal, chi tiết

Microphone

→ Buffer

→ Windown Hanning

→ FFT

→ Gain

→ IFFT

→ Overlap-Add

→ Audio Output



Genre Classification

Audio Input

→ Mono Conversion

→ MFCC Extraction (13 coefficients)

→ Time-Averaged Feature Vector

→ Random Forest Classifier

→ Predicted Genre

→ Confidence Score (predict_proba)

Demo



Environment

- Browser: Chrome / Firefox / Cốc Cốc
- Web Audio API
- Python: 3.10+
- NumPy, SciPy, Librosa, Matplotlib, Scikit-learn
- Web Default Microphone



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SUMMARY

Conclusion

- Successfully implemented FFT-based multi-band Equalizer.
- Implement real-time DSP using Web Audio API.
- Integrated MFCC + Random Forest classification.

Future Develop

- Add more parameters (distortion, reverb, etc.).
- Allow processing multiple tracks at once.
- Real-time waveform transformation.

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Thank You

FOR YOUR ATTENTION

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